Linear Algebra and Its Applications

Section 6.2: Orthogonal Sets

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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6.2 Orthogonal sets

An orthogonal set has pairwise zero inner products; it may include zero. An orthonormal set is orthogonal and has every length equal to 1, so its vectors are necessarily nonzero. For nonzero orthogonal columns u_j, a vector in their span has unique weights $c_j = (x \cdot u_j)/(u_j \cdot u_j)$.

For a line $L = \text{Span}\{u\}$, $u \neq 0$, the orthogonal projection of y is $\widehat{y} = ((y \cdot u)/(u \cdot u))u$. Its residual $y - \widehat{y}$ is perpendicular to the line; the distance to the line is the length of that residual. A rectangular matrix can have orthonormal columns, but the term *orthogonal matrix* is reserved here for a square matrix with $U^{-1} = U^T$.

Practice problem 6.2.P1: Two orthonormal vectors form a plane basis

Their dot product is $(-1/\sqrt{5})(2/\sqrt{5}) + (2/\sqrt{5})(1/\sqrt{5}) = 0$. Their squared lengths are $(1+4)/5 = 1$ and $(4+1)/5 = 1$. Thus they are orthonormal and in particular independent. Two independent vectors in $\mathbb{R}^2$ form a basis.

Conclusion. The proposed pair is an orthonormal basis of $\mathbb{R}^2$.

Practice problem 6.2.P2: The same projection using a different line generator

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 5 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{20}{5} = 4.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The projection is $(8, 4)^T$, the same point obtained with any nonzero generator of the same line.

Practice problem 6.2.P3: A rectangular isometry preserves inner products

The data from Example 6 are

$$U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}, \quad x = (\sqrt{2}, 3)^T, \quad y = (-3\sqrt{2}, 6)^T.$$

Multiplication gives $Ux = (3, -1, 1)^T$ and $Uy = (1, -7, 2)^T$. Consequently $(Ux)\cdot(Uy) = 3+7+2 = 12$, while $x \cdot y = (\sqrt{2})(-3\sqrt{2}) + 3(6) = -6 + 18 = 12$.

Conclusion. The two inner products agree exactly, although the outputs have three coordinates and the inputs only two.

Exercise 6.2.1: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 26 & 0 & 2 \\ 0 & 30 & 0 \\ 2 & 0 & 74 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.2: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 30 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

Exercise 6.2.3: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 54 & 0 & 0 \\ 0 & 126 & -30 \\ 0 & -30 & 11 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.4: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 38 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 56 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

The zero vector causes no difficulty for orthogonality, but it would prevent the set from being independent or orthonormal.

Exercise 6.2.5: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 23 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 122 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

Exercise 6.2.6: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 50 & 0 & 0 \\ 0 & 90 & -32 \\ 0 & -32 & 44 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.7: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^TU = \begin{bmatrix} 13 & 0 \\ 0 & 52 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{39}{13} = 3.$$

$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{26}{52} = \frac{1}{2}.$$

Conclusion. $x = (3)u_1 + (\frac{1}{2})u_2$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.8: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 40 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{-15}{10} = -\frac{3}{2}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{30}{40} = \frac{3}{4}.$$

Conclusion. $x = (-\frac{3}{2})u_1 + (\frac{3}{4})u_2$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.9: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^T U = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{5}{2} = \frac{5}{2}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{-27}{18} = -\frac{3}{2}.$$
$$c_3 = \frac{x \cdot u_3}{u_3 \cdot u_3} = \frac{18}{9} = 2.$$

Conclusion. $x = (\frac{5}{2})u_1 + (-\frac{3}{2})u_2 + (2)u_3$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.10: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^TU = \begin{bmatrix} 18 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 18 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{24}{18} = \frac{4}{3}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{3}{9} = \frac{1}{3}.$$
$$c_3 = \frac{x \cdot u_3}{u_3 \cdot u_3} = \frac{6}{18} = \frac{1}{3}.$$

Conclusion. $x = (\frac{4}{3})u_1 + (\frac{1}{3})u_2 + (\frac{1}{3})u_3$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.11: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 20 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{10}{20} = \frac{1}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -4 \\ 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.12: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 10 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-4}{10} = -\frac{2}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -1 \\ 3 \end{bmatrix} \begin{bmatrix} -\frac{2}{5} \end{bmatrix} = \begin{bmatrix} \frac{2}{5} \\ -\frac{6}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{3}{5} \\ \frac{1}{5} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.13: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 65 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-13}{65} = -\frac{1}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 4 \\ -7 \end{bmatrix} \begin{bmatrix} -\frac{1}{5} \end{bmatrix} = \begin{bmatrix} -\frac{4}{5} \\ \frac{7}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{14}{5} \\ \frac{8}{5} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.14: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 50 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{20}{50} = \frac{2}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} \begin{bmatrix} \frac{2}{5} \end{bmatrix} = \begin{bmatrix} \frac{14}{5} \\ \frac{2}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{4}{5} \\ \frac{28}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.15: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 100 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{100} = \frac{3}{10}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} \begin{bmatrix} \frac{3}{10} \end{bmatrix} = \begin{bmatrix} \frac{12}{5} \\ \frac{9}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{1} = 1$, not the projection length.

Exercise 6.2.16: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 5 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{15}{5} = 3.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -6 \\ 3 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{45} = 3\sqrt{5}$, not the projection length.

Exercise 6.2.17: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}.$$

The set is orthogonal but not normalized. Divide each column by its own positive length to get the following orthonormal columns:

$$Q = \begin{bmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{3}}{3} & 0 \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} \end{bmatrix}, \qquad Q^T Q = I.$$

Exercise 6.2.18: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Conclusion. The set is not orthogonal and hence not orthonormal. Merely normalizing the individual lengths cannot make a nonzero pairwise dot product become zero.

Exercise 6.2.19: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.20: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 \\ 0 & \frac{5}{9} \end{bmatrix}.$$

The set is orthogonal but not normalized. Divide each column by its own positive length to get the following orthonormal columns:

$$Q = \begin{bmatrix} -\frac{2}{3} & \frac{\sqrt{5}}{5} \\ \frac{1}{3} & \frac{2\sqrt{5}}{5} \\ \frac{2}{3} & 0 \end{bmatrix}, \qquad Q^T Q = I.$$

Exercise 6.2.21: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.22: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.23: Orthogonality and orthogonal coordinates

(a) True. For example $(1, 0)^T, (1, 1)^T$ are independent but have dot product 1, so independence need not mean orthogonality.

(b) True. For a nonzero orthogonal family, take $y \cdot u_j$ to obtain $c_j = (y \cdot u_j)/(u_j \cdot u_j)$, without row reduction.

(c) False. Scaling two distinct vectors changes their dot product by a product of scalars; a zero product remains zero. Normalization uses nonzero scales and preserves orthogonality.

(d) False unless the matrix is square. The single column $(1, 0)^T$ is orthonormal but is a rectangular matrix, not an orthogonal matrix.

(e) False. The distance is $\|y - \widehat{y}\|$, not $\|\widehat{y}\|$. For nonzero $y \in L$, the distance is zero while $\|\widehat{y}\| = \|y\| > 0$.

Exercise 6.2.24: Orthogonality and orthogonal coordinates

(a) True. An orthogonal set can contain zero; $\{0, e_1\}$ is orthogonal but dependent. Nonzero orthogonal vectors are independent.

(b) False. The condition gives orthogonality only; the lengths may be different from one. For instance $\{2e_1, 3e_2\}$ meets it.

(c) True. $A^TA = I_n$ implies $\|Ax\|^2 = x^TA^TAx = x^Tx$. Taking nonnegative square roots gives length preservation.

(d) True. The factors of a nonzero c cancel in the projection formula, as proved explicitly in Exercise 31.

(e) True. Invertibility is part of the definition, and its inverse is its transpose.

Exercise 6.2.25: Orthogonality and orthogonal coordinates

Let U have orthonormal columns, so $U^TU = I$.

(b) Prove inner-product preservation first.

For all input vectors x, y, $(Ux) \cdot (Uy) = x^TU^TUy = x^Ty = x \cdot y$.

(a) Deduce preservation of lengths.

Set $y = x$ in the previous identity: $\|Ux\|^2 = \|x\|^2$. Since both norms are nonnegative, $\|Ux\| = \|x\|$.

(c) Deduce preservation of orthogonality in both directions.

Part (b) says that the two dot products are equal. Therefore

$$(Ux) \cdot (Uy) = 0 \quad \Longleftrightarrow \quad x \cdot y = 0.$$

This equivalence includes zero inputs, for which angle formulas would not be defined. As an additional consequence, for nonzero inputs the preserved dot products and norms also preserve the angle cosine.

Conclusion. All three conclusions of Theorem 7 follow from $U^TU = I$.

Exercise 6.2.26: Orthogonality and orthogonal coordinates

A nonzero orthogonal family is independent: in a zero combination, taking the dot product with each generator forces its coefficient to vanish. Thus the given n vectors are independent in $\mathbb{R}^n$. An independent set of exactly n vectors in this n-dimensional space is a basis, so its span is all of $\mathbb{R}^n$.

Conclusion. $W = \mathbb{R}^n$.

Exercise 6.2.27: Orthogonality and orthogonal coordinates

The orthonormal columns of U satisfy $U^TU = I_n$. Their nonzero orthogonality also makes them linearly independent. By the Invertible Matrix Theorem, a square matrix with independent columns is invertible. Multiplying $U^TU = I_n$ on the right by its inverse gives $U^T = U^{-1}$.

Conclusion. U is invertible and orthogonal. Squareness is essential to this inverse conclusion.

Exercise 6.2.28: Orthogonality and orthogonal coordinates

For an orthogonal square matrix, $U^{-1} = U^T$. Hence $UU^T = UU^{-1} = I_n$. The (i, j) entry of UU^T is the inner product of row i and row j. The identity matrix says these products are zero for distinct rows and one for each row with itself. Thus the rows are orthonormal. They are n nonzero orthogonal vectors in $\mathbb{R}^n$, so they form a basis.

Conclusion. The rows form an orthonormal basis, just as the columns do.

Exercise 6.2.29: Orthogonality and orthogonal coordinates

The product of invertible matrices is invertible and $(UV)^{-1} = V^{-1}U^{-1}$. Since U, V are orthogonal,

$$(UV)^{-1} = V^T U^T = (UV)^T,$$

using the transpose-of-product rule in the last equality.

Conclusion. UV is a square invertible matrix whose inverse is its transpose, so it is orthogonal.

Exercise 6.2.30: Orthogonality and orthogonal coordinates

A column permutation keeps the same column vectors, each of length one, with the same zero dot products between distinct columns. Thus V is still square with orthonormal columns. The result of Exercise 27 makes it invertible with inverse V^T. Equivalently write $V = UP$ for a permutation matrix P; then $V^T V = P^T U^T UP = P^T P = I$.

Conclusion. Every column permutation of an orthogonal matrix is orthogonal.

Exercise 6.2.31: Orthogonality and orthogonal coordinates

For $c \neq 0$, replace u by cu in the projection formula:

$$\frac{y \cdot (cu)}{(cu) \cdot (cu)}(cu) = \frac{c(y \cdot u)}{c^2(u \cdot u)}cu = \frac{y \cdot u}{u \cdot u}u.$$

The cancellation is valid because $c \neq 0$ and $u \neq 0$. Any two nonzero generators of a fixed line are related by just such a scale.

Conclusion. The projection depends on the line, not on the chosen nonzero generator or its orientation.

Exercise 6.2.32: Orthogonality and orthogonal coordinates

The two scaled vectors are nonzero because both original vectors and both scalars are nonzero. Their dot product is $(c_1 v_1) \cdot (c_2 v_2) = c_1 c_2 (v_1 \cdot v_2) = c_1 c_2 (0) = 0$. Thus they remain an orthogonal nonzero pair. Applying the same identity to each pair proves the assertion for a whole orthogonal family.

Conclusion. In particular normalization preserves orthogonality.

Exercise 6.2.33: Orthogonality and orthogonal coordinates

For any inputs x, y and scalars a, b,

$$\operatorname{proj}_L(ax+by) = \frac{(ax+by)\cdot u}{u\cdot u}u = a\frac{x\cdot u}{u\cdot u}u + b\frac{y\cdot u}{u\cdot u}u.$$

The right side is $a\operatorname{proj}_L x + b\operatorname{proj}_L y$, so both addition and scalar multiplication are preserved. The fixed nonzero u guarantees the denominator is positive.

Conclusion. Projection onto a fixed line through zero is linear.

Exercise 6.2.34: Orthogonality and orthogonal coordinates

Let $P(y) = \operatorname{proj}_L y$. Exercise 33 proves P linear; the identity map I is also linear. Reflection is $2P - I$. For scalars a, b,

$$(2P-I)(ay+bz) = 2aP(y) + 2bP(z) - ay - bz = a(2P(y)-y) + b(2P(z)-z).$$

Conclusion. Reflection in a fixed line through the origin is linear, with matrix $2uu^T/(u^Tu) - I$.

Exercise 6.2.35: Orthogonality and orthogonal coordinates

The appropriate calculation is the Gram matrix:

$$A^TA = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 100 & 0 & 0 \\ 0 & 0 & 100 & 0 \\ 0 & 0 & 0 & 100 \end{bmatrix}.$$

The six off-diagonal products are all zero, so the four columns are pairwise orthogonal. The diagonal entries also show that each column has length 10.

Conclusion. The columns are orthogonal, and their normalized matrix in the next exercise is $U = A/10$.

Exercise 6.2.36: Orthogonality and orthogonal coordinates

(a) Compute both products with the correct dimensions.

Normalization gives $U = A/10$, so

$$U^TU = I_4, \qquad UU^T = \frac{1}{100}\begin{bmatrix} 82 & 0 & -20 & 8 & 6 & 20 & 24 & 0 \\ 0 & 42 & 24 & 0 & -20 & 6 & 20 & -32 \\ -20 & 24 & 58 & 20 & 0 & 32 & 0 & 6 \\ 8 & 0 & 20 & 82 & 24 & -20 & 6 & 0 \\ 6 & -20 & 0 & 24 & 18 & 0 & -8 & 20 \\ 20 & 6 & 32 & -20 & 0 & 58 & 0 & 24 \\ 24 & 20 & 0 & 6 & -8 & 0 & 18 & -20 \\ 0 & -32 & 6 & 0 & 20 & 24 & -20 & 42 \end{bmatrix}.$$

The first product is 4×4 identity. The second is an 8×8 rank-four projection matrix, not I_8.

(b) Perform one reproducible random-vector calculation.

Integer draws from $[-3, 3]$ with seed 62036 give

$$y = \begin{bmatrix} -1 \\ -1 \\ -1 \\ 0 \\ -1 \\ 2 \\ -3 \\ 0 \end{bmatrix}, \quad p = UU^T y = \begin{bmatrix} -1 \\ -\frac{47}{50} \\ \frac{1}{50} \\ -\frac{11}{10} \\ \frac{1}{5} \\ \frac{29}{50} \\ -\frac{9}{10} \\ \frac{57}{50} \end{bmatrix}, \quad z = y - p = \begin{bmatrix} 0 \\ -\frac{3}{50} \\ -\frac{51}{50} \\ \frac{11}{10} \\ -\frac{6}{5} \\ \frac{71}{50} \\ -\frac{21}{10} \\ -\frac{57}{50} \end{bmatrix}.$$

The formula $p = U(U^T y)$ explicitly writes p as a combination of the columns of U, so $p \in \operatorname{Col} U = \operatorname{Col} A$. Direct multiplication yields $z^T p = 0$.

(c) Check every column, not just p.

$$U^T z = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

These four entries are the four required column dot products.

(d) Extend the check to the whole column space.

If $w = Uc$, then $z \cdot w = z^T Uc = (U^T z)^T c = 0$. Thus $z \in (\operatorname{Col} A)^\perp$, and by construction $y = p + z$.

Conclusion. The displayed vectors give the requested orthogonal decomposition; the orthogonality also follows algebraically from $U^T(y - UU^T y) = 0$.